Tutorial - 5

SVM

suspicious words (x_1)	links (x_2)	Attachment (x_3)	class (spam/Not spam) (y)
6	3	1	+1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1

weighted sum

$$f(x) = z = w_0(b) + w_1 x_1 + w_2 x_2 + w_3 x_3$$

	s.w x_1	Links x_2	Attach x_3
spam →	(4,5,6)	(2,3)	(1)
	$f(x)=0$		
not spam →	(0,1)	(0)	(0)

decision boundary

Intialize : w_2 & $w_3 = 0$

Now $w_1 = 1$, $x_1 = 2$

If $f(x)=0$

$$w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$$

$$w_0 + w_1 x_1 = 0$$

$$w_0 + (1)(2) = 0$$

$$w_0 = -2$$

bias

$$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 - 2$$

$$f(x) = 2 - 2 = 0 < 1$$

$y_i(x_1 w_1 + x_2 w_2 + w_3 x_3 + w_0) \geq 1 \rightarrow$ perfect class

$\leq 1 \rightarrow$ class fail

$$y_i(x_1 - 2) - 6 - 2 = 4$$

$$1(6-2) = 4 \geq 1$$

$$-1(1-2) = 1 \geq 1$$

$$+1(4-2) = 2 \geq 1$$

$$-1(0-1) = 2 \geq 1$$

$$+1(5-2) = 3 \geq 1$$

cost or

loss function $L = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{N} c_i [y_i(wx_i + b) - 1]$

$$= \frac{1}{2}(w_1^2 + w_2^2 + w_3^2) - \sum_{i=1}^{N} c_i \text{ Error}$$

update weights

If P.F $\Rightarrow y_i(w_1x_1 + w_2x_2 + \cdots) \geq 1$

N.P.F $\Rightarrow y_i(w_1x_1 + w_2x_2 + \cdots) < 1$

$$w^{new} = w^{old} - \eta \frac{dL}{dw} \cdot \quad \frac{\partial J}{\partial w} = w \quad J = \frac{1}{2}w^2$$

$$= w - \eta w$$

$$w_1^{new} = w_1^{old} - \eta w_1^{old} = 1 - 0.1(1) = 0.9$$

$$w_0^{new} = w_0^{old} - \eta w_0^{old}$$

$$\text{margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

If N.P.F

$$y_i(w_1x_1 + w_2x_2) < 1 \Rightarrow \frac{2}{\sqrt{(0.9)^2 + 0^2 + 0^2}}$$

$$w^{new} = w - \eta \frac{\partial L}{\partial w}$$

$$w^{new} = w - \eta \frac{\partial L}{\partial w} = \frac{2}{0.9} = 2.222$$

$$= w - \eta \left[w - \sum_i c_i y_i x_i \right]$$

$$= (1-\eta) w + \eta \sum_i c_i y_i x_i$$

$$\sum_i c_i y_i x_i = c \sum_i y_i x_i$$

$C=1 \Rightarrow (y_1x_1 + y_2x_{12} + y_3x_{13} + y_4x_{14} + y_5x_{15})$

$= (1)(6)(-1)(1) + (1)(4) - 1(0) + 1(5)$

If prediction fails you have to show one iteration (epoch) by updating

$$\omega_j^{new} = \omega_j^{old} - \eta \frac{\partial L}{\partial \omega}$$

$$\omega_j^{new} = \omega_j^{old} - \eta \left[\omega - \sum_i c_i y_i x_i\right]$$

$$\omega_1^{new} = \omega_1^{old} - \eta \left[\omega_1^{old} - \sum y_i x_{i1}\right]$$

$$\omega_2^{new} = 1 - 1\left[1 - (-1 \times 3)\right] = -3$$

$$\omega_3^{new} = 1 - 1\left[1 - (-1 \times 1)\right] = -1$$

SVM

$$b^{new} = b^{old} - \eta \frac{\partial L}{\partial b}$$

$$= b^{old} - \eta \left[\sum c_i y_i\right]$$

$$= 0 - 1\left[(1)\cdot(-1)\right] = 1$$

$\omega_1^{new} = -6$

$\omega_2^{new} = -3$

$\omega_3^{new} = -1$

$b^{new} = 1$

$$y_i f(x) = y_i(\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + b)$$

$$= (-6)(1) + (-3)(0) + (-1)(0) + 1$$

$$= -6 + 1$$

$-1 \times 5 = 5$

$y_i f(x) = +5 \geq 1$

$$w_j^{new} = w_j^{old} - \eta \frac{\partial L}{\partial w}$$

If right prediction $L = \frac{1}{2} \|w\|^2$

$$w_j^{old} - \eta w_j^{old}$$

$w_1^{new} = -6 - 1(-6) = 0$

$w_2^{new} = -3 - 1(-3) = 0$

$w_3^{new} = 0$

$$L = \frac{1}{2} \|w\|^2$$

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$b^{new} = b^{old} - \eta \frac{\partial L}{\partial L}$$

$$b^{new} = b^{old} = 1$$

Age (x_1)	Days (x_2)	Lab (x_3)	cost (y)	$\hat{y} \cdot y - \hat{y}$ $(y-\hat{y})^2$
				0.12
2	3	4	12	576 - 556
4	5	6	20	-59256 59286
6	7	8	30	5123322
3	4	5	16	-5123306
5	6	7	27	

Intercept (bias) = b (or) c

$$y = mx + c$$
$$= wx + b$$

predicted : $\hat{y} = c + \underbrace{m_1 x_1 + m_2 x_2 + m_3 x_3}_{xw}$

Intialize : $m_1 = m_2 = m_3 = 0 \quad \eta = 1$ or $0.1 \quad c = 0$

$$\hat{y} = 0 + 0(2) + 0(3) + 0(4) = 0$$
$$y - \hat{y} = 12 - 0 = 12$$

$$m_1^{new} = m_1^{old} + \eta (y - \hat{y}) x_i$$
$$= 0 + 1(12)(2) = 24$$

$$m_2^{new} = 0 + 1(12)(3) = 36 \quad m^3 = 48$$

$$c^{new} = c^{old} + \eta (y - \hat{y})$$
$$= 0 + 1(12) = 12$$

$$\hat{y} = 12 + (24)(4) + (36)(5) + (48)(6)$$
$$= 12 + 96 + 180 + 288$$
$$= 576$$

$$y - \hat{y} = 20 - 576 = -556$$

$$m_1^{new} = m_1^{old} + \eta (y - \hat{y}) x_i$$
$$= 24 + 1(-556)4$$
$$= 24 - 2224 = -2200$$

$m_2^{new} = 36 + 1(-556)5$

$= 36 - 2780$

$= -2744$

$m_3^{new} = 48 + 1(-556)6$

$= 48 - 3336$

$= -3288$

$c^{new} = 12 + 1(-556)$

$= -544$

$\hat{y} = -544 + (-2200)6 + (-2744)7 + (-3288)$

$= -544 - 13200 - 19208 - 26304$

$\hat{y} = -59256$

$m_1^{new} = -2200 + 1(59286)6 = -2200 + 355716$

$= 353516$

$m_2^{new} = -2744 + 1(59286)7 = 412258$

$-2744 + 415002$

$m_3^{new} = -3288 + 1(59286)8 = 471000$

$c^{new} = -544 + 1(59286) = 58742$

$\hat{y} = 58742 + 353516(3) + 412258(4) + 471000(5)$

$= 5123322$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}$$